

Console screenshot

Code for console methods:

```
////////////////////////////////// HOMEWORK 10 CODE ADDITIONS (console) ////////////////////////////////////
// Further code changes found at line 344

// Message Logging examples
console.log("Log Info: App started");
console.info("Info: User opened Giga Pet Game");
console.warn("Warning: This is a demo warning");
console.error("Error: This is a demo error");

console.table([
  { feature: "Celery Button", status: "working" },
  { feature: "Play Button", status: "working" },
  { feature: "Exercise Button", status: "working" },
  { feature: "Treat Button", status: "working" },
  { feature: "Start Button", status: "working" }
]);

console.group("Project 2 Debug Group");
console.log("Current page:", document.title);
console.log("Current URL:", location.href);
console.groupEnd();

console.log("%cCustom Log: Project 2 DevTools Demo", "color: blue; font-size: 18px; font-weight: bold;");

////////////////////////////////// Original Project 2 code ////////////////////////////////////
```

The outputs for the console methods:

The screenshot shows the Chrome DevTools Console with the following content:

- Top bar: Elements, Console (selected), Sources, Network, Performance. 1 error, 1 warning, 1 info.
- Log entries:
 - `{DOMAIN_OF_URL: '127.0.0.1:5500'}` (scriptOnStart.js:1)
 - Log Info: App started (script.js:5)
 - Info: User opened Giga Pet Game (script.js:6)
 - Warning: This is a demo warning (script.js:7)
 - Error: This is a demo error (script.js:8)
- Table (script.js:10):

(index)	feature	status
0	'Celery Button'	'working'
1	'Play Button'	'working'
2	'Exercise Button'	'working'
3	'Treat Button'	'working'
4	'Start Button'	'working'
- Project 2 Debug Group (script.js:18):
 - Current page: Supreme Magnificent Giga Pet (script.js:19)
 - Current URL: <http://127.0.0.1:5500/index.html> (script.js:20)
- Custom Log: Project 2 DevTools Demo (script.js:23) - styled in blue, bold, 18px font.
- Footer: `> [ctrl][i] to turn on code suggestions. Don't show again` (NEW)

Messages logged by the browser

404 message:

- The code

```
// 404 error: Redirect to a page that doesn't exist
function clickedCustomize() {
  if (started) {
    console.log("Loading customization options...");

    fetch("customize-pet-options.json") // file doesn't exist
    | | .catch(err => console.error("Customization failed:", err));
  }
}
```

-
- This function is tied to the “Customize Pet” button
 - What it does is try to fetch a non-existent file for customizing the user’s pet. Instead of just leading to a 404 page, instead it logs the error in the console.

- The console screenshot



TypeError message:

- The code

```
// Type error: undefined "fly" and activate causes type error
function clickedFlyButton() {
  if (started) {
    console.warn("Pet tried to use a nonexistent ability (flight)...");

    // Play bite sound
    $('#flight-audio')[0].play();

    pet_info.fly.activate();
  }
}
```

-
- This function is tied to the “Make Pet Fly” button
 - The function attempts to call a method on an undefined property “fly” leading to a TypeError, a runtime error that is thrown when operations are performed on an invalid type
- The console screenshot

```
▶ {DOMAIN_OF_URL: '127.0.0.1:5500'} scriptOnStart.js:1
▲ ▶ Pet tried to use a nonexistent ability (flight)... script.js:373
✖ ▶ Uncaught TypeError: Cannot read properties of undefined (reading 'activate') script.js:378
  at HTMLButtonElement.clickedFlyButton (script.js:378:22)
  at HTMLButtonElement.dispatch (jquery-2.2.1.min.js:3:7516)
  at n.event.add.r.handle (jquery-2.2.1.min.js:3:5597)
> | ctrl i to turn on code suggestions. Don't show again NEW
```

○

Violation message

- The code

```
// Cause a violation
function clickedNeutronStar() {
  if (started) {
    console.warn("Overfeeding pet... this may cause performance issues!");

    const startTime = Date.now();

    // Play bite sound
    $('#star_explode')[0].play();

    // Run for ~3 seconds (long task)
    while (Date.now() - startTime < 5000) {
      // add weight continuously
      weight += 10000;
    }

    console.log("Finished feeding a neutron star");
    checkAndUpdatePetInfoInHtml();
  }
}
```

○

- The function is tied to the “Feed Neutron Star” button

- The function creates a violation because of a long-running task in the function (3 second of running the task)

- The console screenshot (with verbose enabled to allow the error message to show)

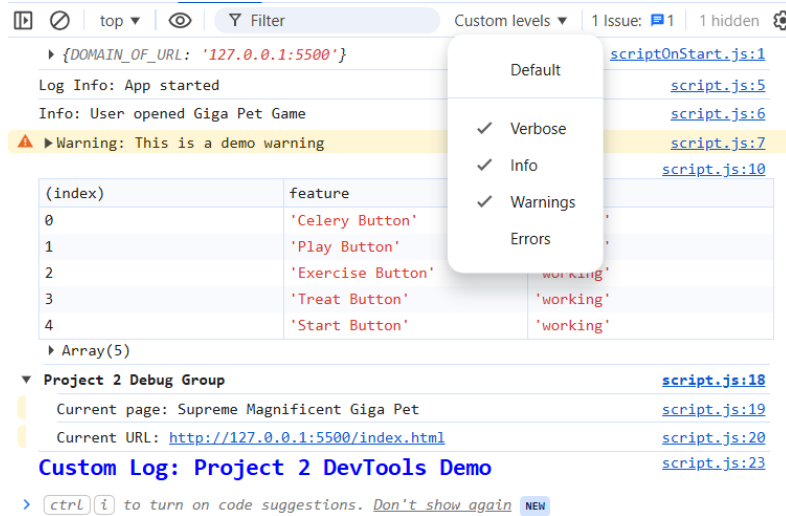
```
▲ ▶ Overfeeding pet... this may cause performance issues! script.js:352
Finished feeding a neutron star script.js:365
[Violation] 'click' handler took 5000ms jquery-2.2.1.min.js:3
> | ctrl i to turn on code suggestions. Don't show again NEW
```

○

Filter Messages

Filter by log level

- Filter so error messages don't appear



○

Filter by text

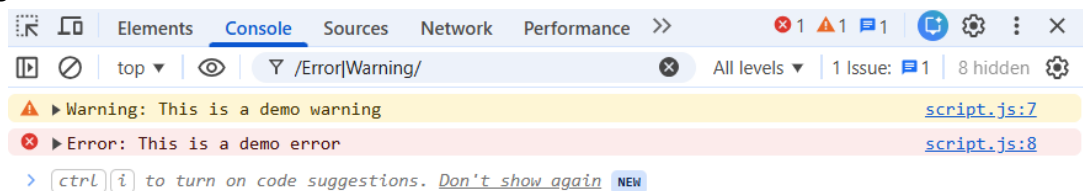
- In console, I filter only on messages that have the text “demo”



○

Filter by regular expressions

- Here I use the regular expression “/Error|Warning/” to only show messages that contain a warning or error



○

Filter by message source

- Here I open the side bar and filter by message source. Specifically I am looking at the messages from the script.js file

Log Info: App started [script.js:5](#)

Info: User opened Giga Pet Game [script.js:6](#)

Warning: This is a demo warning [script.js:7](#)

Error: This is a demo error [script.js:8](#)

(index)	feature	status
0	'Celery Button'	'working'
1	'Play Button'	'working'
2	'Exercise Button'	'working'
3	'Treat Button'	'working'
4	'Start Button'	'working'

Array(5)

Project 2 Debug Group [script.js:18](#)

Current page: Supreme Magnificent Giga Pet [script.js:19](#)

Current URL: <http://127.0.0.1:5500/index.html> [script.js:20](#)

Custom Log: Project 2 DevTools Demo [script.js:23](#)

> [ctrl](#) [i](#) to turn on code suggestions. Don't show again [NEW](#)

Filter by user messages

- Here I filter by user messages

{DOMAIN_OF_URL: '127.0.0.1:5500'} [scriptOnStart.js:1](#)

Log Info: App started [script.js:5](#)

Info: User opened Giga Pet Game [script.js:6](#)

Warning: This is a demo warning [script.js:7](#)

Error: This is a demo error [script.js:8](#)

(index)	feature	status
0	'Celery Button'	'working'
1	'Play Button'	'working'
2	'Exercise Button'	'working'
3	'Treat Button'	'working'
4	'Start Button'	'working'

Array(5)

Project 2 Debug Group [script.js:18](#)

Current page: Supreme Magnificent Giga Pet [script.js:19](#)

Current URL: <http://127.0.0.1:5500/index.html> [script.js:20](#)

Custom Log: Project 2 DevTools Demo [script.js:23](#)

> [ctrl](#) [i](#) to turn on code suggestions. Don't show again [NEW](#)

Bug fix:

For the function feedKibble(), which is tied to the Feed Kibble button, we have:

```
function feedKibble() {
  if (started) {
    // Intentionally create a weight addition value as a string
    var weightAddition = "10";

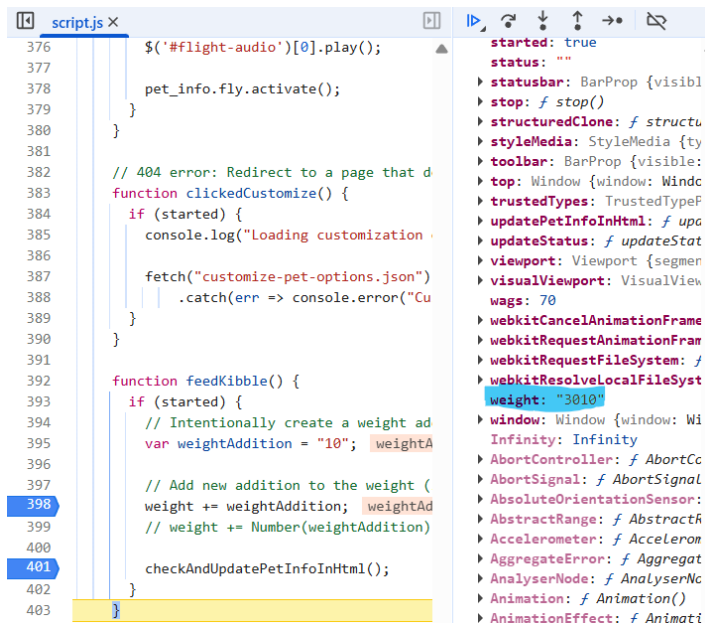
    // Add new addition to the weight (if not fixed, will concatenate instead)
    weight += weightAddition;

    checkAndUpdatePetInfoInHtml();
  }
}
```

When the Feed Kibble button is pressed (after the start button has been pressed), the result shows (the weight was initially 30 lbs):

Name: Pet Name
Weight: 3010 pounds
Happiness: 70 tail wags (per min)
Energy: 40 %

In the sources tab when debugging, after the completion of the feedKibble() function, I see that the value of weight is wrapped in quotes (I also create breakpoints on lines 398 and 401 and show the value of the weight variable in the scope pane):



And when using watch expressions on both the weight and weightAddition variables, we see their types, and that they are both strings:

```
376     $('#flight-audio')[0].play();
377
378     pet_info.fly.activate();
379 }
380
381
382 // 404 error: Redirect to a page that d
383 function clickedCustomize() {
384     if (started) {
385         console.log("Loading customization .
386
387         fetch("customize-pet-options.json")
388             .catch(err => console.error("Cu
389     }
390 }
391
392 function feedKibble() {
393     if (started) {
394         // Intentionally create a weight ad
395         var weightAddition = "10"; weightA
396
397         // Add new addition to the weight (
398         weight += weightAddition; weightAd
399         // weight += Number(weightAddition)
400
401         checkAndUpdatePetInfoInHtml();
402     }
403 }
```

Debugger paused

Watch

- typeof weight: "string"
- typeof weightAddit... : "string"

- Breakpoints
- Scope
- Call Stack
- XHR/fetch Breakpoints
- DOM Breakpoints
- Global Listeners
- Event Listener Breakpoints
- CSP Violation Breakpoints

However, instead of giving the pet kibble, for example with giving a treat, we see that the type is supposed to be a number:

```
179     if (isTransitioning) {
180         return; // ignore click
181     }
182
183     if (!petLeft && !overweight) {
184         energy += 20;
185         weight += 8;
186         wags -= 13;
187
188         // Play bite sound
189         $('#bite-audio')[0].play();
190         checkAndUpdatePetInfoInHtml();
191     }
192 }
193
194 function checkAndUpdatePetInfoInHtml()
195     // Make any necessary changes to not
196     checkWeightAndHappinessBeforeUpdating
197
198     // Update attributes
199     pet_info.weight = weight;
200     pet_info.happiness = wags;
201     pet_info.energy = energy;
202
203     updateStatus();
204     updatePetInfoInHtml();
205 }
206
```

Debugger paused

Watch

- typeof weight: "number"

- Breakpoints
- Scope
- Call Stack
- XHR/fetch Breakpoints
- DOM Breakpoints
- Global Listeners
- Event Listener Breakpoints
- CSP Violation Breakpoints

To fix this, we can convert weightAddition to a number and doing the addition, which gives us the correct result in the console:

```
> weight += Number(weightAddition)
< 40
>
```

So back to the code, we fix this:

```
function feedKibble() {
  if (started) {
    // Intentionally create a weight addition value as a string
    var weightAddition = "10";

    // Add new addition to the weight (if not fixed, will concatenate instead)
    weight += Number(weightAddition); // fix

    checkAndUpdatePetInfoInHtml();
  }
}
```

And now we get the correct result, where weight actually becomes 40:

The screenshot shows the Giga Pet Game interface on the left and the browser's developer tools on the right. The game interface displays a pet's stats: Name: Pet Name, Weight: 40 pounds, Happiness: 70 tail wags (per min), and Energy: 40 %. There are buttons for Treat, Play, Exercise, Feed Celery, RESET, Feed Neutron Star, Make Pet Fly, Customize Pet, and Feed Kibble. A pet image is shown with a speech bubble saying "Hello! I would like to play and eat." The developer tools on the right show the Sources panel with the script.js file open. A breakpoint is set at line 398, and the console shows the log output: "Loading...". The Watch panel shows the current state of the weightAddition variable as a string.